

8. Write a program to implement Linear Regression (LR) algorithm in python

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

from sklearn.metrics import mean_squared_error, r2_score

X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)

y = np.array([2, 4, 5, 4, 5, 7, 8, 9, 10, 12])

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Slope (Coefficient):", model.coef_[0])

print("Intercept:", model.intercept_)

print("Predicted values:", y_pred)

print("Actual values:", y_test)

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)

print("R2 Score:", r2)

plt.scatter(X, y, label="Actual Data")

plt.plot(X, model.predict(X), label="Regression Line")

plt.xlabel("X")

plt.ylabel("Y")

plt.title("Linear Regression")
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plt.legend()
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plt.show()
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```
Slope (Coefficient): 1.0689655172413794  
Intercept: 0.6206896551724128  
Predicted values: [10.24137931  2.75862069]  
Actual values: [10  4]  
Mean Squared Error: 0.7996432818073731  
R2 Score: 0.9111507464658475
```

